

### General Instructions:

- The submission due date of this assignment is **Thursday (30<sup>th</sup> April 2026) midnight** (before **12:00 am**)
- Write a report (i.e. in a word file) that illustrates your main solution steps including the best fitness values and the average values, plotted over generations (with and without elitism)
- Zip your code and the report in a file entitled [**YourName\_YourID\_AssignmentNumber**]
- This assignment should be delivered and discussed INDIVIDUALLY .

### ➤ Requirements:

**In this assignment it is required to implement a Simplex search (Nelder-Mead) to solve the following problem:**

We are minimizing

$$f(x, y) = (x - 2)^2 + (y - 2)^2$$

The Algorithm for Simplex can be described as follows:

1. Starting with an initial simplex, the algorithm determines the worst  $X_w$ , second worst  $X_s$  and the best  $X_b$  vertices of the current simplex.
2. To determine a new point, the centroid  $X_c$  of all the other points except the worst point is calculated  $X_c = \frac{X_b + X_g}{2}$  and point  $X_w$  is reflected as  $X_r = X_c + \alpha(X_c - X_w)$

\* Outside contraction (when  $f_{\text{good}} \leq f(r) < f_{\text{worst}}$ ):

$$c_{\text{out}} = \text{centroid} + \rho(r - \text{centroid}),$$

\* Inside contraction (when  $f(r) \geq f_{\text{worst}}$ ):

$$c_{\text{in}} = \text{centroid} + \rho(\text{worst} - \text{centroid})$$

$$A = (0, 0),$$

$$B = (1, 0),$$

$$C = (0, 1),$$

3. Create Python Code, *using  $\beta, \alpha$  and  $\gamma$  random*